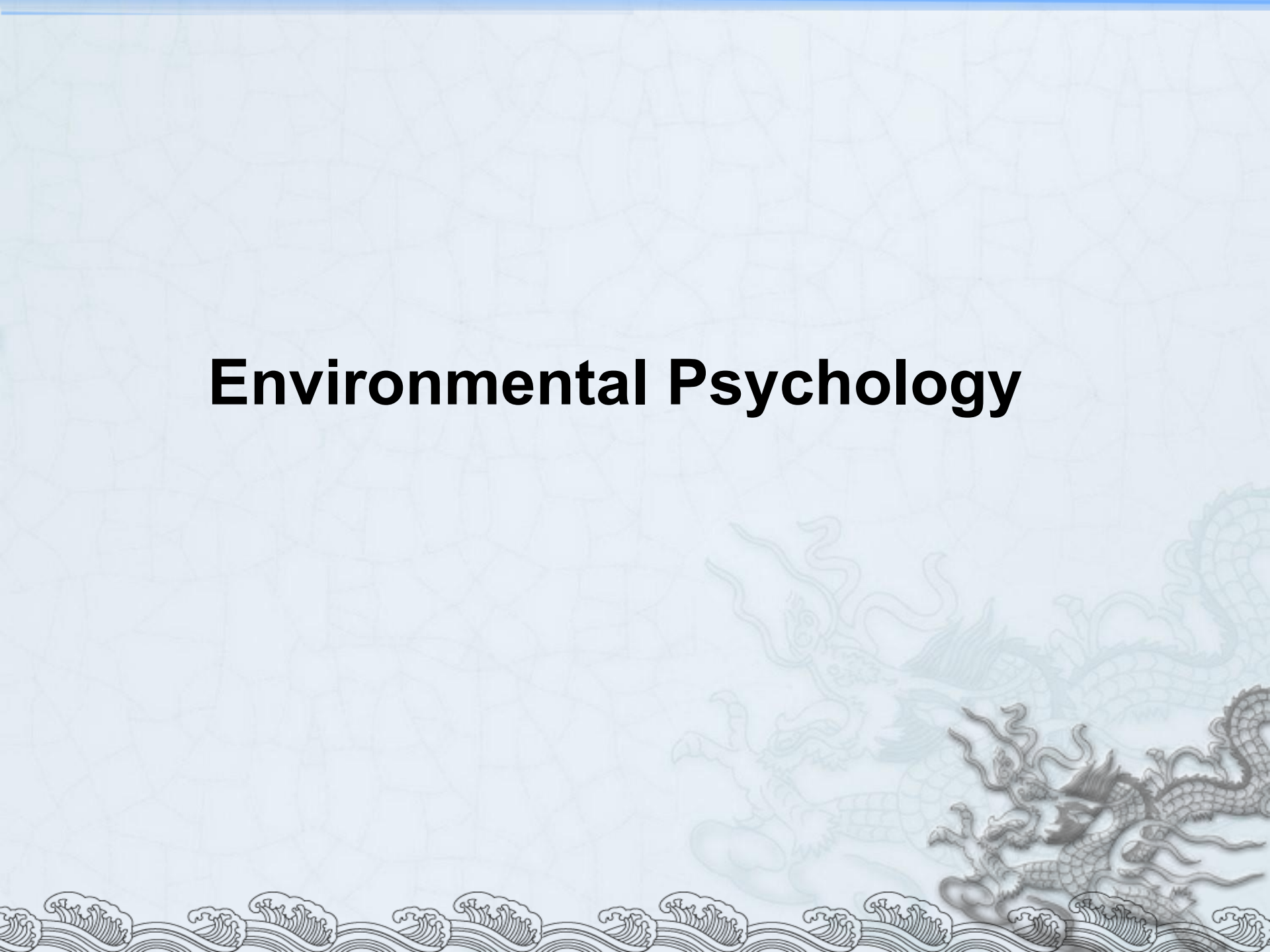
Environmental Psychology



Lecture three

Perception & Cognition

Part One

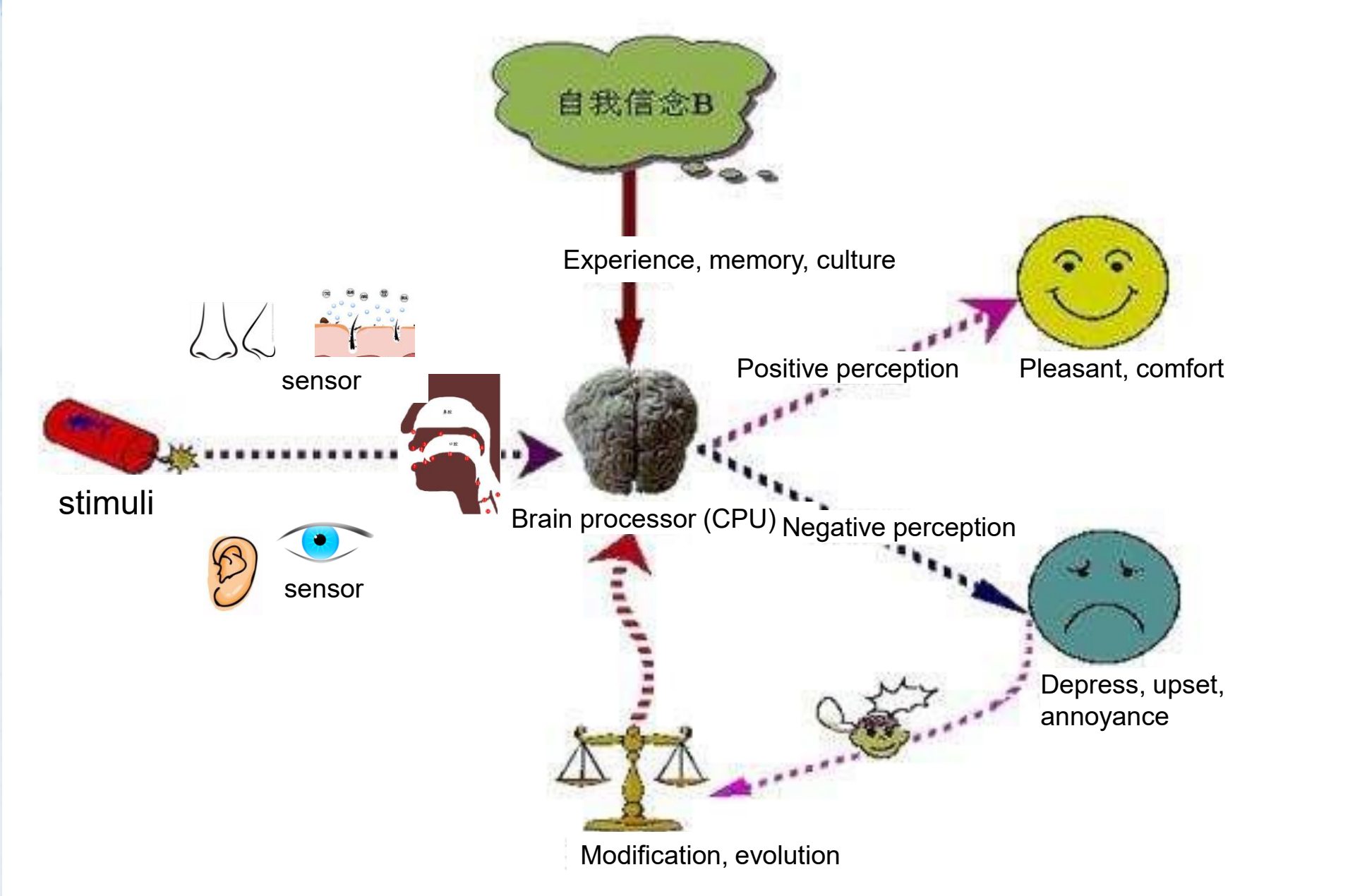
- Environmental perception
- Mind, behavior, action

Part Two

- Environmental cognition



Feel → Scare → run


自我信念B
Experience, memory, culture
sensor
Positive perception
Pleasant, comfort
stimuli
Brain processor (CPU)
Negative perception
sensor
Depress, upset, annoyance
Modification, evolution

Environmental Perception

- Thinking about what we can do in a perceiving environment, as well as visual, auditory, and other imagery of the scene – cognitive.
- Environmental perception includes valuation, or the determination of good and bad elements – evaluative.
- Our feelings about the environment influence our perception of it – affective.
- Environmental perception encompasses the meaning we derive from an environment – interpretive.

The term environmental perception sometimes is used more broadly to include aspects of how we appraise and assess environments, it will primarily refer to the initial information – gathering phase of the process.

- Environmental perception is the initial gathering of information. We primarily are visual beings, but environmental perception includes the ways and means by which we collect information through all our senses.


Air, lighting, etc.
Feel → want to do


Building, water, people
Feel → want to do

Environmental Perception

- Environmental perception includes cognitive, affective, interpretive, and evaluative components, all operating at the same time.

Indeed, the affective and evaluative components of environmental perception are the roots of forming attitudes towards the environment.





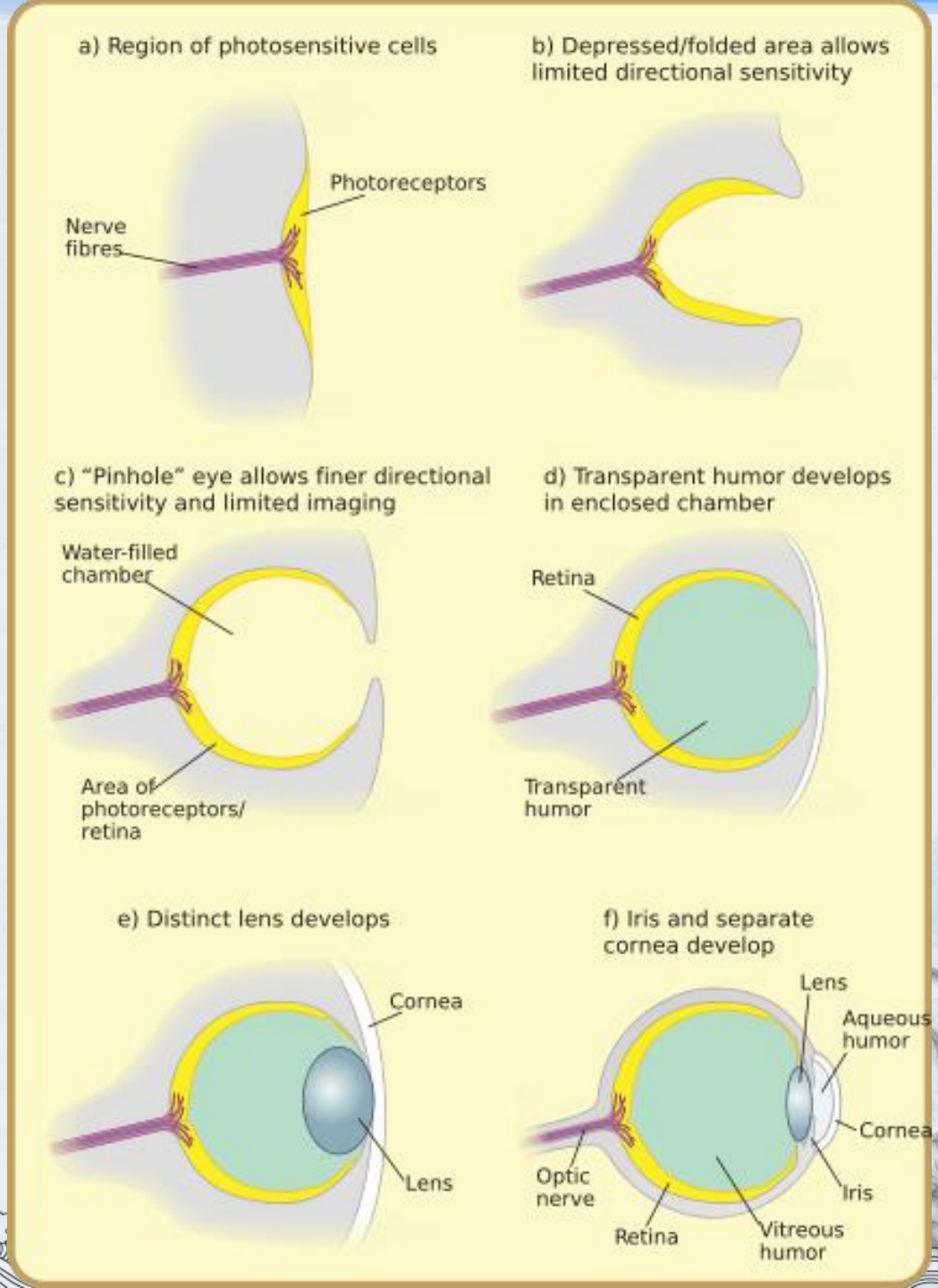
Noise made by a dog

Actions taken by noise - Control strategy; Noise barrier

- The stimulus-response perceptual relationship between an urban landscape and an urban habitant.

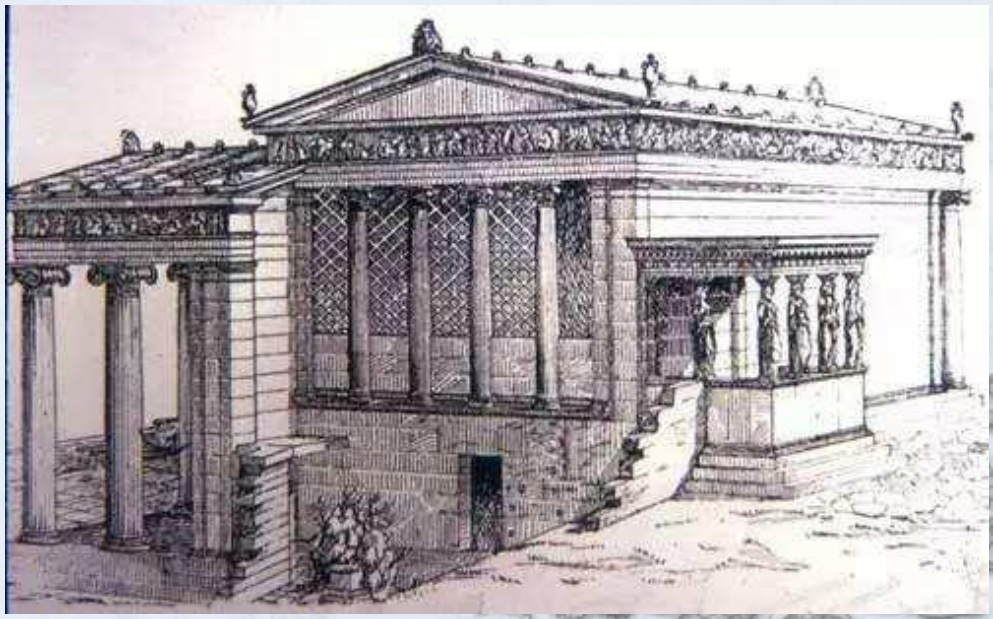
  - Pattern or object?
  - Color or texture?
  - Flower or road?
  - Light or atmosphere?




a) Region of photosensitive cells
Photoreceptors
Nerve fibres
b) Depressed/folded area allows limited directional sensitivity
c) "Pinhole" eye allows finer directional sensitivity and limited imaging
Water-filled chamber
Area of photoreceptors/ retina
d) Transparent humor develops in enclosed chamber
Retina
Transparent humor
e) Distinct lens develops
Cornea
Lens
f) Iris and separate cornea develop
Lens
Aqueous humor
Cornea
Iris
Optic nerve
Retina
Vitreous humor

- Environmental psychology looks upon the stimulus and its perception as a unit that contains more than just a stimulus and a response.
- It depends not just upon the individual stimuli in the landscape, but also depends upon the patterning, complexity, novelty, and movement of the contents of the landscape and upon the past experience of the perceiver, his or her ability to impose structure on the landscape, his or her auditory and olfactory associations with the landscape, and his or her personality characteristics.





Environmental Perception

- The difference between the research of environmental perception and traditional perception is not merely in the size and complexity of the stimulus presented; the role of the perceiver is also different. In environmental perception studies, the participants often move around in and through the environmental display means that the perceiver experiences it from multiple perspectives.









What is Environmental Perception

- For such differences, it becomes obvious that our two imaginary visitors have different backgrounds.
- What is curious, is that the actual physical scenery is **identical** for both of them.
- The environment is the same, but the perceptual experience is vastly different.
- Even though the environments to which they return are quite different, there will be many similarities in their perceptual experiences.

- A stimulus is an event in the environment external to the organism.
- The organism becomes aware of these stimuli by means of receptor nerve cells, which are sensitive to specific forms of energy, such as light, sound, or temperature.
- When these energy sources are sufficiently strong to stimulate the receptor cells, we refer to the process as sensation.
- When a number of sensations are put together by mediational processes in the brain so that we recognize or organize a pattern out of these sensations, the process is **called perception.**

Environmental Perception

- To the individual from Shanghai, **Lanzhou might appear rather simple**, lacking a central core of skyscrapers and large industrial and commercial districts, whereas the individual from Xizhang would probably evaluate **Lanzhou as large**, with a complex pattern of different styles of buildings.

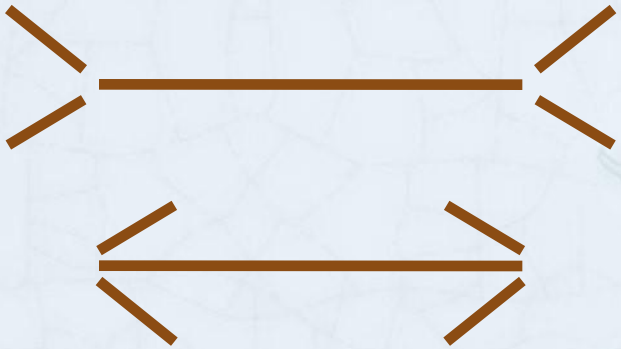
- What are the psychological factors that result in these two individuals' different perceptions and evaluations?

- We will begin by looking at how we perceive, through experience, both individual aspects of the environment and collective or holistic aspects.

- The factors involved in hazard perception, as well as sociocultural influences on perception.

Social and cultural influences on environmental perception

- The Müller-Lyer illusion: To those of us raised in built environment, the line at the top appears longer than the one on the bottom.



Social and cultural influences on environmental perception

- Social and culture factors, such as sex roles, socioeconomic status, and exposure to modern architecture, influence what one learns or what one has the opportunity to experience, it stands to reason that such factors influence how one learns to perceive the environment.

- ***Social influences on air pollution***: in general, high socioeconomic level residents who live in relatively unpolluted suburban areas are less concerned about pollution than are lower socioeconomic level citizens who are more likely to live near sources of pollution.
- The most aware of air pollution are higher socioeconomic level individuals who live in areas with high levels of pollution.
- Baker offers an explanation for this perceptual difference by noting that when high pollution levels exist, lower socioeconomic groups either are primarily concerned with more immediate problems of day-to-day living or believe they are helpless to fight the problem of air pollution.

Perception of change

- If we readily adapt to environmental stimulation, how do we perceive change in such things as air pollution and urban blight?
- If we live in an area where air pollution is high, and we adapt to this pollution, how can we perceive changes in level of pollution?
- The intensity of a new stimulus required for it to be perceived as different from the present stimulus is proportionate to the present stimulus.
- This function generally applies to all forms of stimulation, including light, sound, pressure, and smell.
- For example, a community with little pollution should become alarmed when massive clouds of brown smog suddenly appear, but large urban areas with heavy smog should require extremely high levels of additional pollution before becoming alarmed.
- One oil derrick on an undisturbed seashore is noticed immediately, but one derrick added to a field of 50 probably goes unnoticed.

When we change an environment, as building a residential subdivision on farmland to afford us shelter, the immediate environment no longer readily affords life support for wild animals or for the farmer. Perception of this environment, according to Gibson, depends on the gain and loss of affordances for each organism.



Movement, Habituation, Adaptation

- By far, we have talked about perception without regard time.
- ***Perception of movement:*** At the receptor level, perception of movement is instigated by successive stimulation of neighboring receptors.
- We feel bugs crawling on our skin because bugs trigger a series of pressure receptors as they crawl along.
- ***Habituation:*** referring to a physiological process.
- Cognitive explanations of habituation emphasize a cognitive reappraisal of the stimulus as less deserving of attention after repeated presentation.
- The first time you hear a loud noise, you allocate considerable attention to…..once you know …….you probably evaluate it as nonthreatening….the next time….
- It is fairly easy to adapt to a constant hum in the background or to a smell that constantly permeates the immediate atmosphere – ***Adaptation.***
- It is much more difficult to adapt to the irregular noise or a smell that blows in with sporadic gusts of wind – ***Adaptation.***

Environmental cognition

- Even when we are not actively viewing, hearing, or smelling an environment, we can still experience it mentally and determine something about our location in it from this mental experience. sensory – perceptual system.
- Mental representation or cognition about the layout is termed a cognitive map.
- Cognitive maps present for us whether we are in the mapped environment or somewhere else. – way finding.
- The concept of cognitive mapping has its roots in the work of *E.C. Tolman*, who described the way rats learn to "map" the environment of an experimental maze.
- Work by *Lynch* gave the impetus to studies of humans' cognitive maps of cities, and more recent work has expanded our knowledge of mapping techniques in many environments.

- The general way of thinking about, recognizing and organizing a layout is termed ***environmental cognition – space syntax*** encompasses a set of theories and techniques for the analysis of spatial configurations, as a tool to help architects simulate the likely social effects of their designs.
- Originally it was conceived by Bill Hillier, Julienne Hanson at the Bartlett, UCL.

Components of cognitive maps

- Lynch found that five categories of features could be used to describe and analyze cognitive maps: paths, edges, districts, nodes, and landmarks



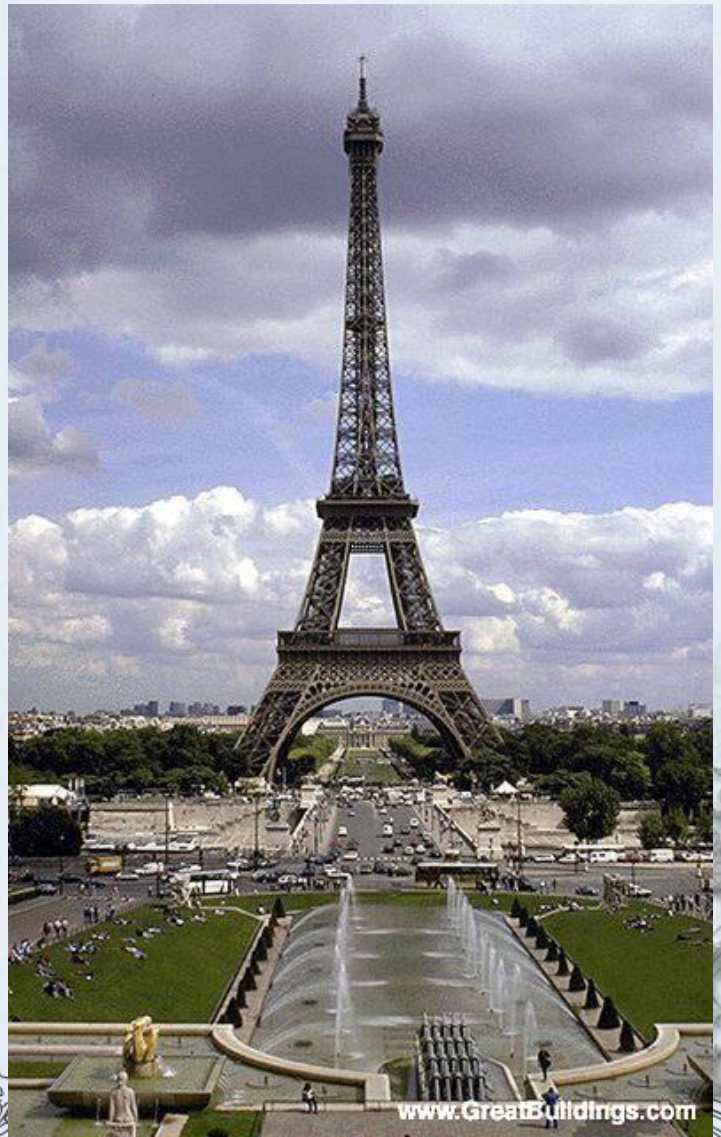






Components of cognitive maps

- Lynch found that five categories of features could be used to describe and analyze cognitive maps: paths, edges, districts, nodes, and landmarks



Landmarks



Cognition maps

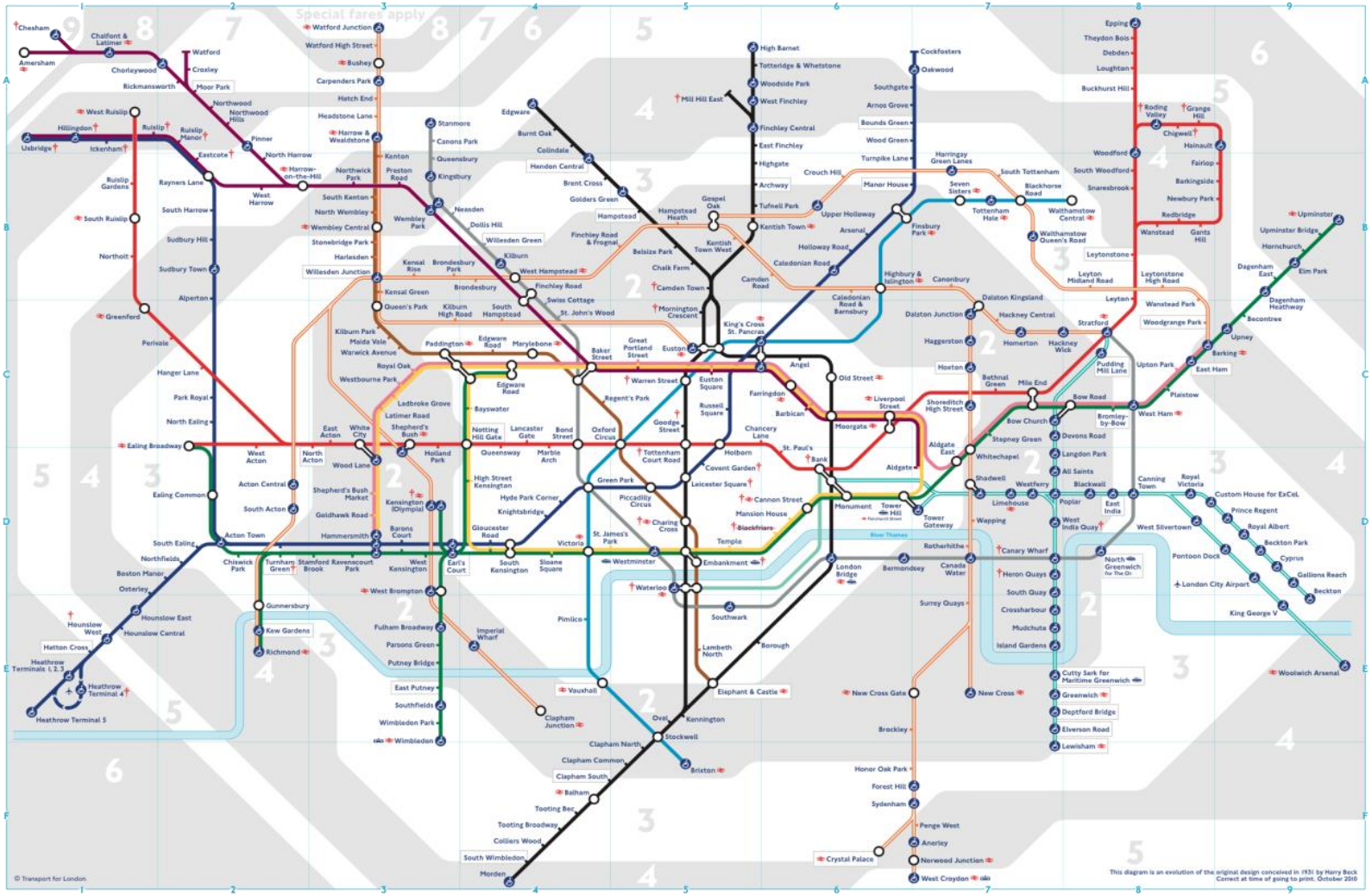
- Environmental cognition concerns the way we acquire, store, organize, and recall information Evidence of cognitive mapping is in advertisements, subway maps, magazines, and in every person's memory and thinking.
- People do not process information about the environment the way cameras or computers do.

- Spatial cognition includes the concept of cognitive maps, which are pictorial and semantic images in our heads of how places are arranged.
- Evidence of cognitive mapping is in advertisements, subway maps, magazines, and in every person's memory and thinking
- From a mechanical point of view, our processing information from our environment is full of errors.
- Our cognitions also differ from one person to the next.

- What is a cognitive map of a city? Cognitive maps are “cities of the mind” that we carry around in our heads, representing the city as we know it.
- A cognitive map is juxtaposed with an aerial map of a city.
- Compared to “reality” the cognitive map is sketchy, incomplete, and distorted.
- Cognitive map is sketchy, impressionistic, almost like a projective test reflecting our views of the city.

London tube map : *sketchy map*

Tube map



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Transport for London

UNDERGROUND



Functions of cognitive maps

- Familiarity and socioeconomic class, gender:
- the more familiar we are with an environment, the more accurate and detailed are our cognitive maps.
- Upper-class individuals have the more thorough cognitive maps of an environment is likely to be.
- Man seems to draw a more thorough cognitive maps.
- Cognitive maps serve the adaptive functions of letting us get around in our environment efficiently and of helping us locate valuable items within the environment.
- Without a cognitive map, we would have to search for locations in our environment in a haphazard manner.
- Cognitive maps are used to communicate locations to others and to understand others' communications about location to us.

End of the Third Lecture

Thanks for attention

Any questions? (leilayu@hit.edu.cn)